

CERTIFICATE OF ANALYSIS 409603

Client Details

Client	Macquarie Geotech (Sydney)
Attention	Jasper Haines
Address	3 Watt Dr, Bathurst, NSW, 2795

Sample Details

Your Reference	<u>S26240-1, Chowder Bay DFI (60740478)</u>
Number of Samples	10 Soil
Date samples received	25/05/2026
Date completed instructions received	25/05/2026

Analysis Details

Please refer to the following pages for results, methodology summary and quality control data.

Samples were analysed as received from the client unless as indicated below in the method summaries. Results relate specifically to the samples as received.

Results are reported on a dry weight basis for solids and on an as received basis for other matrices.

Report Details

Date results requested by	01/06/2026
Date of Issue	28/05/2026

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Accredited for compliance with ISO/IEC 17025 - Testing. **Tests not covered by NATA are denoted with ***

Results Approved By

Jenny He, Inorganic Team Leader

Authorised By

Nancy Zhang, Laboratory Manager

Misc Inorg - Soil

Our Reference		409603-1	409603-2	409603-3	409603-4	409603-5
Your Reference	UNITS	S116199	S116200	S116201	S116202	S116203
Sample ID		HA2_0.4-0.8	HA3_0.4-0.8	ABH2_1.0-1.2	ABH5_0.13-1.5	ABH8_0.48-1.1
Date Sampled		24/03/2026	24/03/2026	24/03/2026	24/03/2026	24/03/2026
Type of sample		Soil	Soil	Soil	Soil	Soil
Date prepared	-	27/05/2026	27/05/2026	27/05/2026	27/05/2026	27/05/2026
Date analysed	-	27/05/2026	27/05/2026	27/05/2026	27/05/2026	27/05/2026
pH 1:5 soil:water	pH Units	4.9	4.7	7.8	6.5	6.2
Electrical Conductivity	µS/cm	90	74	130	92	53
Chloride, Cl 1:5 soil:water	mg/kg	28	30	<10	10	31
Sulphate, SO4 1:5 soil:water	mg/kg	110	48	85	90	39
Resistivity in soil*	ohm m	110	140	76	110	190

Misc Inorg - Soil

Our Reference		409603-6	409603-7	409603-8	409603-9	409603-10
Your Reference	UNITS	S116204	S116205	S116206	S116207	S116208
Sample ID		ABH2_2-3	ABH3_2.0-3.0	ABH4_1.5-2	ABH4_6.0-6.45	ABH6_0.5-1.5
Date Sampled		24/03/2026	24/03/2026	24/03/2026	24/03/2026	24/03/2026
Type of sample		Soil	Soil	Soil	Soil	Soil
Date prepared	-	27/05/2026	27/05/2026	27/05/2026	27/05/2026	27/05/2026
Date analysed	-	27/05/2026	27/05/2026	27/05/2026	27/05/2026	27/05/2026
pH 1:5 soil:water	pH Units	5.2	7.8	8.7	6.9	8.0
Electrical Conductivity	µS/cm	42	83	84	180	110
Chloride, Cl 1:5 soil:water	mg/kg	<10	21	<10	56	<10
Sulphate, SO4 1:5 soil:water	mg/kg	53	46	20	190	29
Resistivity in soil*	ohm m	240	120	120	56	93

Method ID	Methodology Summary
Inorg-001	pH - Measured using pH meter and electrode. Please note that the results for water analyses are indicative only, as analysis outside of the APHA storage times.
Inorg-002	Conductivity and Salinity - measured using a conductivity cell.
Inorg-002	Conductivity and Salinity - measured using a conductivity cell at 25oC in accordance with APHA 22nd ED 2510 and Rayment & Lyons. Resistivity is calculated from Conductivity (non NATA). Resistivity (calculated) may not correlate with results otherwise obtained using Resistivity-Current method, depending on the nature of the soil being analysed.
Inorg-081	Anions - a range of Anions are determined by Ion Chromatography, in accordance with APHA latest edition, 4110-B. Waters samples are filtered on receipt prior to analysis. Alternatively determined by colourimetry/turbidity using Discrete Analyser.

QUALITY CONTROL: Misc Inorg - Soil						Duplicate			Spike Recovery %	
Test Description	Units	PQL	Method	Blank	#	Base	Dup.	RPD	LCS-1	[NT]
Date prepared	-			27/05/2026	2	27/05/2026	27/05/2026		27/05/2026	[NT]
Date analysed	-			27/05/2026	2	27/05/2026	27/05/2026		27/05/2026	[NT]
pH 1:5 soil:water	pH Units		Inorg-001	[NT]	2	4.7	4.8	2	102	[NT]
Electrical Conductivity	µS/cm	2	Inorg-002	<2	2	74	69	7	106	[NT]
Chloride, Cl 1:5 soil:water	mg/kg	10	Inorg-081	<10	2	30	24	22	110	[NT]
Sulphate, SO4 1:5 soil:water	mg/kg	10	Inorg-081	<10	2	48	46	4	113	[NT]
Resistivity in soil*	ohm m	1	Inorg-002	<1	2	140	150	7	[NT]	[NT]

Result Definitions

NT	Not tested
NA	Test not required
INS	Insufficient sample for this test
PQL	Practical Quantitation Limit
<	Less than
>	Greater than
RPD	Relative Percent Difference
LCS	Laboratory Control Sample
NS	Not specified
NEPM	National Environmental Protection Measure
NR	Not Reported

Quality Control Definitions

Blank	This is the component of the analytical signal which is not derived from the sample but from reagents, glassware etc, can be determined by processing solvents and reagents in exactly the same manner as for samples.
Duplicate	This is the complete duplicate analysis of a sample from the process batch. If possible, the sample selected should be one where the analyte concentration is easily measurable.
Matrix Spike	A portion of the sample is spiked with a known concentration of target analyte. The purpose of the matrix spike is to monitor the performance of the analytical method used and to determine whether matrix interferences exist.
LCS (Laboratory Control Sample)	This comprises either a standard reference material or a control matrix (such as a blank sand or water) fortified with analytes representative of the analyte class. It is simply a check sample.
Surrogate Spike	Surrogates are known additions to each sample, blank, matrix spike and LCS in a batch, of compounds which are similar to the analyte of interest, however are not expected to be found in real samples.

Laboratory Acceptance Criteria

Duplicate sample and matrix spike recoveries may not be reported on smaller jobs, however, were analysed at a frequency to meet or exceed NEPM requirements. All samples are tested in batches of 20. The duplicate sample RPD and matrix spike recoveries for the batch were within the laboratory acceptance criteria.

Filters, swabs, wipes, tubes and badges will not have duplicate data as the whole sample is generally extracted during sample extraction.

Spikes for Physical and Aggregate Tests are not applicable.

For VOCs in water samples, three vials are required for duplicate or spike analysis.

Duplicates: >10xPQL - RPD acceptance criteria will vary depending on the analytes and the analytical techniques but is typically in the range 20%-50% – see ELN-P05 QA/QC tables for details; <10xPQL - RPD are higher as the results approach PQL and the estimated measurement uncertainty will statistically increase.

Air volumes are typically provided by customers (often as flow rate(s) and sampling time(s) and/or simply volumes) sampled or exposure times (determines 'volume' passive badges are exposed to)). Hence in such circumstances the volume measurement is inevitably not covered by Envirolab's NATA accreditation. An exception may occur where Envirolab Newcastle does the sampling where accreditation exists for certain types of sampling and hence volume determination(s). Note air volumes are often used to determine concentrations for dust and/or analyses on filters, sorbents and in impingers. For canister sampling, the air volume is covered by Envirolab's NATA accreditation.

Where air volumes and/or sampled areas and/or time based sampling data is supplied by customers and the customer is NATA accredited for the sampling activity then calculated concentrations can be reported on the endorsed Certificate of Analysis (CoA). Confirmation of the accreditation must be supplied by the customer and such information is used in good faith. Where the sampling is not covered by NATA accreditation then only the raw data can be reported on the CoA (e.g. mass, fibres per field).

Matrix Spikes, LCS and Surrogate recoveries: Generally 70-130% for inorganics/metals (not SPOCAS); 60-140% for organics/SPOCAS (+/-50% surrogates) and 10-140% for labile SVOCs (including labile surrogates), ultra trace organics and speciated phenols is acceptable.

In circumstances where no duplicate and/or sample spike has been reported at 1 in 10 and/or 1 in 20 samples respectively, the sample volume submitted was insufficient in order to satisfy laboratory QA/QC protocols.

When samples are received where certain analytes are outside of recommended technical holding times (THTs), the analysis has proceeded. Where analytes are on the verge of breaching THTs, every effort will be made to analyse within the THT or as soon as practicable.

Where sampling dates are not provided, Envirolab are not in a position to comment on the validity of the analysis where recommended technical holding times may have been breached.

Measurement Uncertainty estimates can be downloaded from the [Envirolab Resources website](#) or obtained directly by contacting the laboratory.

Analysis of aqueous samples typically involves the extraction/digestion and/or analysis of the liquid phase only (i.e. NOT any settled sediment phase but inclusive of suspended particles if present), unless stipulated on the Envirolab COC and/or by correspondence. Notable exceptions include certain Physical Tests (pH/EC/BOD/COD/Apparent Colour etc.), Solids testing, total recoverable metals and PFAS where solids are included by default.

For Dust Deposit Gauge (DDG) analysis the sampling, sampling period and funnel exposure area do not fall under Envirolab's NATA accreditation (unless the Newcastle laboratory where responsible for the sampling), hence the annotation on the DDG units of reporting.

Urine Analysis - The BEI values listed are taken from the 2022 edition of "TLVs and BEIs Threshold Limits" by ACGIH.